

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech. Branch : Computer Science and Engineering & Allied Semester : 5

Subject Code & Name: BTCOC503 & Software Engineering

Max Marks: 60 Date: 11/02/2025 Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.

2. Question No. 1 will be compulsory and include objective-type questions.

3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.

4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.

5. Use of non-programmable scientific calculators is allowed.

6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which one of the following activities would span all stages of a software development life cycle (SDLC) in waterfall-based software development?	Remember	1
	a. Coding b. Testing c. Project management d. Design		
2	Which of the following are not the characteristic of an agile software development project?	Remember	1
	a. Shared code ownership b. Implement the simplest solution to meet today's problem c. Continual feedback from customer d. Postpone integration until all the incremental deliveries are complete		
3	Pair programming intends to take which one of the following best practices to the extreme?	Remember	1
	a. Design b. Incremental development c. Code review d. Simplicity		

4	Consider the following partial requirement for a software: "It should be possible to use a web browser as the front end." It can be considered to be which one of the following types of requirements?	Remember	1
	a. Functional requirement b. Non-functional requirement c. Design constraint d. External interface		
5	Which one of the following most accurately defines the term software process model?	Remember	1
	a. The way in which software is developed b. The way in which software processes data c. The way in which software is used d. The way in which software behaves		
6	Which one of the following phases accounts for the maximum effort during development of a typical software product?	Remember	1
	a. Coding b. Testing c. Designing d. Specification		
7	Which one of the following phrases would most appropriately complete the sentence: "The later a defect is found during a traditional software development cycle _____."	Remember	1
	a. the less troublesome it is b. the more expensive it is to repair c. the less important it is to the product's operation d. the more important it is to the product's operation		
8	Which one of the following statements is true concerning unit testing?	Remember	1
	a. Carried out by the development team b. Involves testing the system as a whole c. Often carried out by a separate testing team d. Carried out by the customers		
9	Which one of the following types of bugs may not get detected in black-box testing, but are very likely to be get detected by white-box testing?	Remember	1

	a. Incorrect functionality b. Missing functionality c. Trojans d. Incorrect algorithm		
10	Which of the following is not a part of Software evolution?	Remember	1
	a. Re-engineering activities b. Maintenance activities c. Development activities d. Negotiating with client		
11	Which of the following UML diagrams has a static view?	Remember	1
	a. Collaboration b. Use case c. State chart d. Activity		
12	Which diagram in UML shows a complete or partial view of the structure of a modeled system at a specific time?	Remember	1
	a. Sequence Diagram b. Collaboration Diagram c. Class Diagram d. Object Diagram		
Q. 2	Solve the following.		12
A)	Explain the software Engineering code of ethics and professional practice. https://www.batuonline.com	CO1	6
B)	Explain software process activities.	CO1	6
Q. 3	Solve the following.		12
A)	Discuss the Extreme Programming process.	CO2	6
B)	What is agile Software Development? Give its advantages.	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	What is system model? Explain types of system models in brief.	CO3	6

B)	What are the Architectural patterns? Explain Model view controller (MVC) pattern	CO3	6
C)	Explain in brief i) Context Models and ii) Interaction models	CO3	6
Q.5	Solve Any Two of the following.		12
A)	What is software reuse? Explain software reuse levels.	CO4	6
B)	Draw sequence diagram for ATM.	CO4	6
C)	Discuss Object Oriented Design using UML in brief.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Explain test driven development with neat sketch.	CO5	6
B)	Compare software inspection and testing	CO5	6
C)	Discuss Component testing in detail.	CO5	6
	*** End ***		

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